MIL-I-46557A (MU)
30 OCTOBER 1963
SUPERSEDING
MIL-I-46557 (ORD)
28 JULY 1961

MILITARY SPECIFICATION

IGNITER, PERCUSSION-ELECTRIC, M30 PARTS FOR

1. SCOPE

1.1 This specification covers the metal parts for one type of igniter designated as Igniter, Percussion-Electric, M30.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-2550 —Ammunition and Special Weapons; General Specification for

MIL-P-10025 —Packaging, Packing and Marking for Interplant Shipment of Inert Ammunition Components, General Specification for

MIL-I-45208 —Inspection Requirements, General Specification for

MIL-I-45607 —Inspection Equipment, Supply and Maintenance of

STANDARDS

MILITARY

MIL-STD-105—Sampling Procedures and Tables for Inspection by Attributes

MIL-STD-109—Quality Assurance Terms and Definitions

MIL-STD-1235—Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

DRAWINGS

ORDNANCE CORPS

7548481—Igniter, Percussion-Electric, M30 for Artillery Primers, Body

7548482—Igniter, Percussion-Electric, M30 for Artillery Primers, Plug, Closing

7548483—Disc, Closing

PUBLICATIONS

ORDNANCE CORPS

IEL-7156122—Index of Inspection Equipment Lists

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

3. REQUIREMENTS

3.1 **Material.** Materials shall be in accord-

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ance with applicable drawings and specifications.

3.2 Parts. The parts shall comply with all requirements specified on Drawings (dwgs.) 7548481, 7548482 and 7548483 as applicable and with all requirements specified in applicable specifications.

3.3 Workmanship. All parts shall contain required chamfers and, be dry and free of chips, burs, dirt, grease, rust and other foreign material. The cleaning method shall not be injurious to any of the parts nor shall the parts be contaminated by the cleaning agents.

4. QUALITY ASSURANCE PROVISIONS

4.1 General quality assurance provisions. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements. Reference shall be made to Standard MIL–STD–109 in order to define terms used herein. The provisions of Specification MIL–A–2550 shall apply.

4.1.1 *Contractor quality assurance system.* The contractor shall provide and maintain an effective quality assurance system in compliance with the requirements of Specification MIL–I–45208.

4.1.2 *Submission of product.* At the time the completed lot of product is submitted to the Government for acceptance the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

(a) A statement that the lot complies with all quality assurance provisions of the approved current written description of the system.
(b) Number of units of product inspected.
(c) Results obtained for all inspection performed.
(d) Drawing, specification number and date, together with an identification and date of changes.
(e) Certificates of analysis on all material purchased by the contractor when such material is controlled by Government specifications referenced in any of the contractual documents.
(f) Number of items in the lot.
(g) Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.1.3 *Government verification.* Using the contractor's written quality assurance procedure, this detail specification, the applicable drawings and other contracual documents as a guide, the Government inspector shall verify at unscheduled intervals all quality assurance operations performed by the contractor. Verification will be in accordance with Specification MIL–I–45208 and will be performed to the extent necessary to assure compliance with the contractual requirements. Severity of Government inspection of individual characteristics will be directly related to the seriousness of the classification assigned. In no instance will a characteristic classified "critical" be accepted solely on the basis of the contractor's records.

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4.2 Initial production inspection.

4.2.1 Prior to the start of regular production, the contractor shall submit an initial production sample to a Government approved facility as designated by the Contracting Officer (see 6.4) for evaluation in accordance with the provisions of 4.2.2. The initial production sample shall consist of 25 complete sets of parts which have been produced by the contractor or furnished by the supplier and which have been manufactured using the same production processes, procedures and requirement which will be used in fulfilling the contract. All parts and materials, including packaging and packing, shall be obtained from the same sources of supply as will be used in regular production. The sample shall be accompanied by certificates of analysis for materials. An initial production sample, or portion thereof, as directed by the Contracting Officer, shall also be submitted whenever there is a lapse in production for a period in excess of 90 days, or whenever a change occurs in manufacturing process, material used, drawing, specification or source of supply as to significantly affect product uniformity as determined by the Government. Prior to submission, the contractor shall inspect the sample to the degree necessary to assure that it conforms to the requirements of the contract and submit a record of this inspection with the sample. A sample containing known defects will not be submitted unless specifically authorized by the Contracting Officer.

4.2.2 *Inspections to be performed.* Each igniter component shall be inspected for any or all of the requirements of this specification and for any or all of the requirements of the applicable drawings.

4.2.3 *Rejection.* If any component fails to comply with any or all of the applicable requirements, the initial production sample shall be rejected. The Government reserves the right to terminate its inspection upon any failure of a component in the sample to comply with any or all of the stated requirements. In the event of rejection, the Government reserves the right to require the contractor to take corrective action and submit a new initial production sample or portion thereof. Until an initial production sample is accepted, the contractor is, in no way, authorized by the Government to initiate regular production unless otherwise directed by the Contracting Officer.

4.3 Inspection provisions.

4.3.1 *Lot formation.* A lot shall consist of parts produced by one manufacturer in one unchanged process, in accordance with the same drawing, same drawing revision, same specification, and same specification revision. Drawing, specification and process changes not affecting safety, performance, or fit, as determined by the Government, shall not necessitate changing the lot interfix number.

4.3.1.1 Closing discs (see dwg. 7548483). The sampling plan for the inspections for "diameter", "thickness" and "ragged edge" shall be in accordance with Standard MIL–STD–105, table IV B, Code Letter H and an AQL of 1.5 percent.

4.3.2 *Examination.* Sampling plans and procedures for the following classifications of defects shall be in accordance with Standard MIL–STD–105 except that inspection for critical defects, when listed, shall be 100 percent. Continuous sampling plans in accordance with Standard MIL–STD–1235 may be used if approved by the procuring activity. Also, at the option of the procuring activity, AQL's and sampling plans may be applied to the individual characteristics listed using an AQL of 0.25 percent for each major defect and an AQL of 0.40 percent for each minor defect. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.3.3.

4.3.2.1 Body (see dwg. 7548481).

Categories	*Defects*	*Method of inspection*	*Code No. (see 6.2)*
Critical:	None defined		
Major:	AQL 0.65 percent		
101.	Pitch diameter of external thread, minimum (min.)	Gage	01001
102.	Major diameter of external thread, min.	Gage	01002

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Categories	Defects	Method of inspection	Code No. (see 6.2)
	103. Pitch diameter of internal thread, maximum (max.)	Gage	01003
	104. Minor diameter of internal thread, max.	Gage	01004
	105. Diameter of body, max.	Gage	01005
Minor:	AQL 6.50 percent		
	201. Pitch diameter of external thread, max.	Gage	01006
	202. Major diameter of external thread, max.	Gage	01007
	203. Pitch diameter of internal thread, min.	Gage	01008
	204. Length of internal thread, min.	Gage	01009
	205. Length of external threads	Gage	01010
	206. Diameter of ignition element cavity	Gage	01011
	207. Diameter of hole through base	Gage	01012
	208. Diameter of flange	Gage	01013
	209. Concentricity of ignition element cavity with external threads	Gage	01014
	210. Concentricity of ignition element hole with ignition element cavity	Gage	01015
	211. Length of thread undercut, max.	Gage	01016
	212. Location of wrench holes	Gage	01017
	213. Total length	Gage	01018
	214. Thickness through bottom of ignition element cavity	Gage	01019
	215. Thickness of flange	Gage	01020
	216. Depth to chamfered surface inside	Gage	01021
	217. Any operation missing	Visual	01022
	218. Wrench hole depth inadequate	Visual	01023
	219. Evidence of poor workmanship (see 3.3)	Visual	01024
	220. Chamfer at hole through base at interior, excessive (shoulder missing)	Visual	01025

4.3.2.2 Plug, closing (see dwg. 7548482).

Categories	Defects	Method of inspection	Code No.
Critical:	None defined		
Major:	AQL 0.40 percent		
	101. Pitch diameter of thread, min.	Gage	02001
	102. Major diameter of thread, min.	Gage	02002
	103. Flash hole missing or obstructed	Visual	02003
Minor:	AQL 4.0 percent		
	201. Pitch diameter of thread, max.	Gage	02004
	202. Major diameter of thread, max.	Gage	02005
	203. Diameter of hole	Gage	02006
	204. Diameter and angle of countersink	Gage	02007
	205. Total length	Gage	02008
	206. Slot missing	Visual	02009
	207. Finish not as required	Visual	02010
	208. Evidence of poor workmanship (see 3.3)	Visual	02011

4.3.3 *Inspection equipment.* Index of Inspection Equipment Number IEL 7156122 identifies the inspection equipment required to perform the examinations and tests prescribed in this section. The contractor shall design inspection equipment in accordance with the instructions in paragraph 6.3.

4.3.3.1 *Government rights to documentation.* Inspection equipment drawings and lists provided and revised in accordance with the requirements of the IEL may be used by DOD activities for design, procurement, manufacture, testing, evaluation, production and receiving inspection, overhaul, shipping, storage, identification of stock, ordering and storage of replacement parts, inspection of items at overhaul, general maintenance of equipment, construction, survey and whenever inspection equipment drawings are needed.

4.3.3.2 *Supply and maintenace.* Supply and maintenance of the equipment listed on the IEL shall be in accordance with Specification MIL–I–45607.

4.3.3.3 *Government use of contractor's inspection and test equipment.* The contractor shall make available to the Government all inspection and test equipment necessary for determining conformance with contract requirements. Personnel for operating the equipment, and verification of its accuracy, shall be supplied by the contractor for the performance of examination or test by the Government.

5. PREPARATION FOR DELIVERY

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5.1 Packing and marking.

5.1.1 *Level C.* The parts shall be packed and marked in accordance with Specification MIL–P–10025.

5.2 Data cards. Data cards shall be prepared for each lot in accordance with the information specified in Specification MIL–A–2550.

6. NOTES

6.1 Ordering data. Procurement documents shall specify the following:

(a) Title, number and date of this specification.

6.2 Inspection code numbers. The five digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government.

6.3 Inspection equipment. The contractor shall design inspection equipment as required by the referenced Inspection Equipment Lists (IEL) in accordance with the instructions of 6.3.1 through 6.3.7.

6.3.1 *Inspection equipment lists.* Inspection equipment lists indicate the availability of inspection equipment designs by showing in the "number" column of the list of inspection equipment (00 Form 1242–3) the numbers of drawings of existing equipment designs or codes as indicated in 6.3.2. Design action required of the contractor with respect to the different types of drawings that may be listed is described in 6.3.3 and 6.3.4. Action required by the contractor with respect to commercial inspection equipment is described in 6.3.5. The contractor will be required to prepare detailed drawings in accordance with 6.3.6 for all the equipment coded as "contractor design" in the number column. These contractor designs must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.7.

6.3.2 *Inspection equipment list codes.* The inspection equipment as defined in 6.3.3, 6.3.4, 6.3.5, and 6.3.6 will be designated in the Inspection Equipment List by the following codes:

CDOF —Contractor's design responsibility on Ordnance format; ORDM–608–12.

CDCF —Contractor's design responsibility on contractor format.

OD —Ordnance design.

ODMU—Ordnance design, mandatory for use.

CE —Commercial equipment.

SCD —Specification Control Drawing.

6.3.3 *Ordnance designs.* Ordnance designs are reflected on detailed drawings which completely depict all the information necessary for the fabrication of the item of inspection equipment. The contractor need provide no design when an Ordnance design is listed for an item of inspection equipment. Ordnance designs fall into two basic classifications; mandatory and nonmandatory. When an inspection equipment list references mandatory Ordnance designs, the contractor shall comply with, and use these designs accordingly. The contractor may, however, in connection with nonmandatory designs and with the approval of the Government, design alternate inspection equipment or use comparable commercial equipment to facilitate his operations. Such contractor prepared designs or commercial equipment selections must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.7.

6.3.4 *Specification control drawings.* Specification control drawings depict the minimum equipment requirements in outline, descriptive, diagrammatic, or pictorial form only and specify the required performance or other characteristics. Contractors must prepare detailed drawings (see 6.3.6) of their designs in support of specification control drawings. These contractor prepared designs must be approved by the Government prior to the fabrication or procuring of the equipment. Commercial equipment meeting the requirements of specification control

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drawings may be approved if described in sufficient detail to permit identification and evaluation by the Government. Designs shall be submitted for approval as specified in 6.3.7.

6.3.5 *Commercial equipment.* Commercial equipment is inspection equipment that has universal application for a specific function. It is comprised of items commonly used by industry and government. Contractors are not required to furnish drawings of commercial inspection equipment but a list of such equipment must be approved by the Government. Lists shall be submitted for approval to the inspection element of the agency administering the contract.

6.3.6 *Contractor designs.* Contractor designs are designs of inspection equipment for which the Government has assigned design responsibility to the contractor. Contractor designs shall be supported by detailed drawings which depict all information necessary to completely fabricate, calibrate and operate an item of inspection equipment. This requires that the necessary views, dimensions, materials, finish, notes, operating and calibration instructions be properly depicted in accordance with approved practices to the extent that further calculation or clarification will not be required. Contractor designs identified as CDCF may be developed on the format the contractor normally employs in his equipment design procedure provided such format reflects the detail and information specified above. Contractor designs identified as CDOF shall comply with the format and requirements of ORDM 608–12, and, in addition, contain the detail and information specified above.

6.3.7 *Submission of contractor designs.* Designs shall be submitted for approval to the Commanding Officer, Picatinny Arsenal, ATTN: SMUPA–ND. Design review will normally be accomplished within one month after receipt by Picatinny Arsenal. Partial submission of inspection equipment designs is permissible and encouraged. However, the Arsenal completion date for design review will be based on the date of the final submission of designs.

6.4 Submission of initial production samples. Instructions as to the location for evaluation of initial production shall be obtained from the contracting officer. Upon receipt of such request, the contracting officer shall advise Picatinny Arsenal and instructions will be issued accordingly. All inquiries should be forwarded to: Commanding Officer, Picatinny Arsenal, ATTN: SMUPA–NI.

Notice. When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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